Actual prescribed quotient-buffer capacity

November9,2026. Source `8e62d4dfc`. Unsigned producer evidence [pcc:unverified].

Exact accounting. Let B count all prescribed coefficient slot widths, one quotient vector per actual node. For $0 < u \le a$ and $h = \lceil 3M/u \rceil$, Lean identifies the actual-node sum with

$$B = \sum_{i=0}^{u-1} 2^{a-i} h(u-i)\big[p + 8h2^{a-i} + 8M(a+1)\big].$$

Actual valuation cardinalities supply each level count. The zero node and inactive levels have zero order. Keeping only the odd-node level gives

$$B \ge 8h^2 u 4^a, \qquad uB \ge 72M^2 4^a.$$

The second inequality uses the actual source ceiling $hu \ge 3M$.

Meaning of the charge. Simultaneous fixed-width packed storage of all those prescribed slots charges at least B bits. This is a capacity accounting convention, including slots whose value is zero. It is not an information-theoretic lower bound against streaming, sparse or compressed representations, shared storage, alternate precision schemes or other algorithms. No RSS, allocation or timing measurement has been taken.

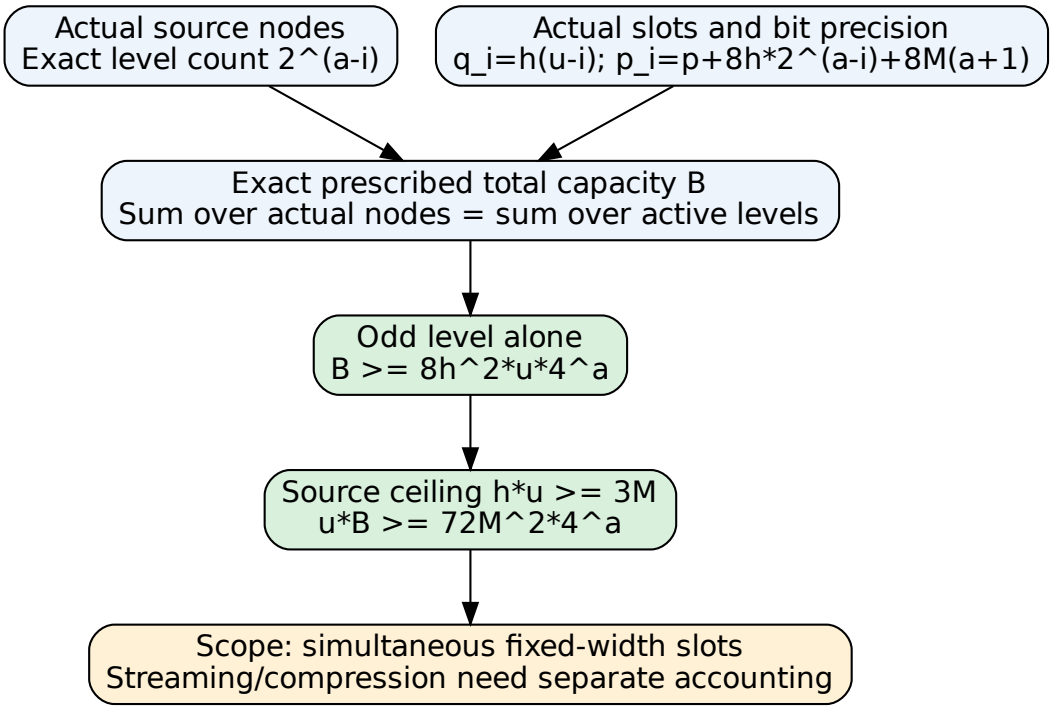
Parameter implication. For input length n, a linear grid exponent a makes this simultaneous capacity exponential through 4^a, since $u \le a$. A sub-exponential version of this explicit strategy needs a sublinear grid exponent or a representation avoiding simultaneous capacity. This condition is necessary for this strategy and gives no sufficient recursive runtime theorem. The manuscript uses $\lambda = \lceil \log_2 s \rceil$ and $a \le 4\epsilon\lambda$, hence $L \le s^{1/1000}$. Its intended $4^a = L^2$ factor is polynomial in s; this bound does not reject that schedule.

Auxiliary algebra. The formal weighted sum $W_r(u) = \sum_{j=0}^{u-1}(j+1)r^j$ satisfies

$$(r-1)^2 W_r(u) = [u(r-1) - 1]r^u + 1.$$

Specializations $r = 2, 4$ are available; connecting a fully closed form for B is a separate next step.

Dependency map and evidence



Source bindings. `BufferBudget.actualBufferBitsEq`, `actualBufferBitsLower`, `actualBufferBitsPortLower` and `weightedIdentity`. Their canonical PCC pins and final audit/replay receipts are linked by the companion Markdown report. Failed build002 is retained; successful builds003/004 contain no unsolved declarations.

Sources and continuation. Actual node counts, source orders and child precision from the preceding formal recovery component; pinned manuscript D.9/D.10 as cited there. KB unavailable (wrapper has no executable). Native census protocol is prospectively frozen; S3 admission failed for missing transport and no data was collected. Continue parameter/cost derivation while that experiment-only dependency is resolved. Streaming needs separate retained state and recomputation bounds.